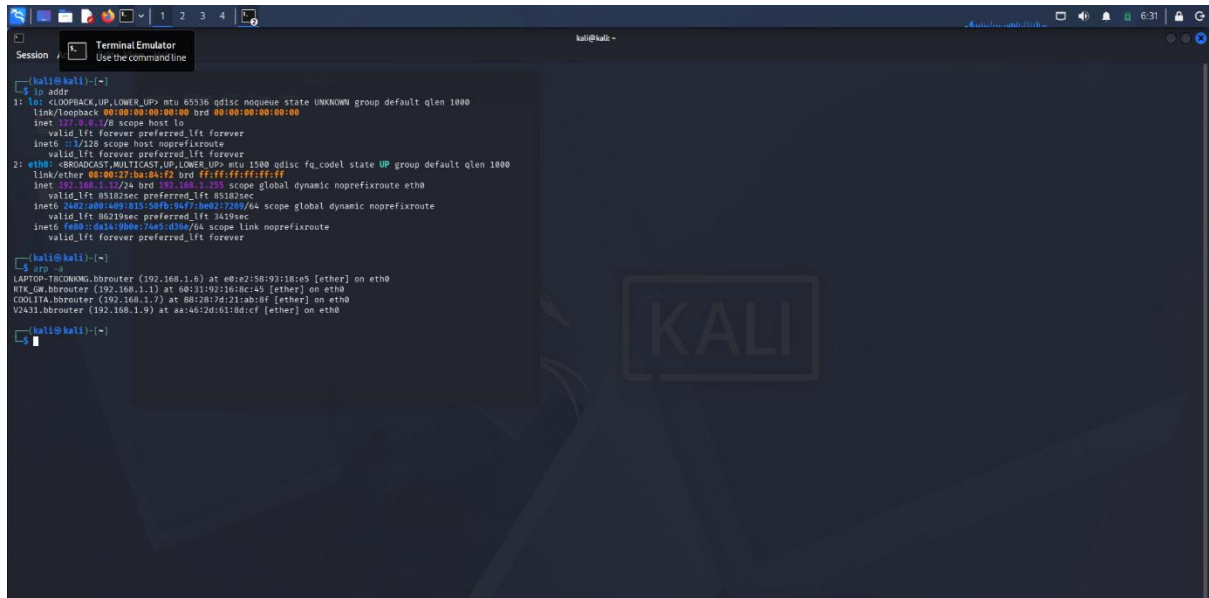


1. Install Nmap on your system and use it to scan your local Wi-Fi network for all connected devices. List the IP addresses and device names (if available) found in the scan.

```
kali@kali:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:00:27:ba:04:f2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.12/24 brd 192.168.1.255 scope global dynamic noprefixroute eth0
        valid_lft 86349sec preferred_lft 86349sec
    inet6 2402:a00:1409:815:50fb:94f7:b002:7269/64 scope global dynamic noprefixroute
        valid_lft 86349sec preferred_lft 3549sec
    inet6 fe80::d814:900e:74e5:d30e/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

```
kali@kali:~$ sudo nmap -sn 192.168.1.0/24
[sudo] password for kali:
Starting Nmap 7.90 ( https://nmap.org ) at 2026-09-16 06:15 -0400
Nmap scan report for RTK_QW.bbrouter (192.168.1.1)
Host is up (0.0030s latency).
MAC Address: 60:31:92:16:8c:45 (DNT India pvt)
Nmap scan report for LAPTOP-T8CONKMG.bbrouter (192.168.1.6)
Host is up (0.00068s latency).
MAC Address: E0:E2:58:93:18:ES (Intel Corporate)
Nmap scan report for COOLITA.bbrouter (192.168.1.7)
Host is up (0.088s latency).
MAC Address: 88:28:70:21:AB:8F (AltoBean (China))
Nmap scan report for V2431.bbrouter (192.168.1.9)
Host is up (0.081s latency).
MAC Address: AA:46:20:61:80:CF (Unknown)
Nmap scan report for Kali.bbrouter (192.168.1.12)
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 4.06 seconds
```

2. Use the 'arp -a' command in your terminal or command prompt to display the current ARP table. Identify at least two devices on your network and write down their IP and MAC addresses.



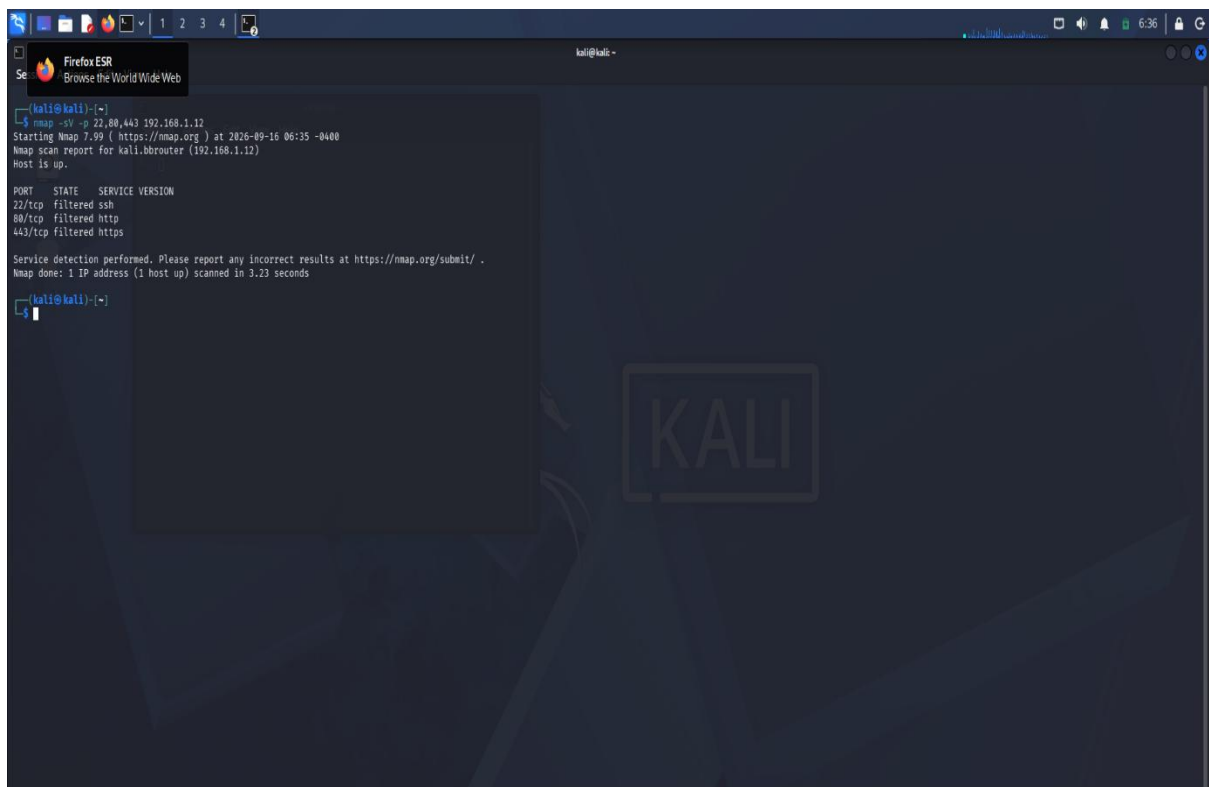
The terminal shows the configuration of a loopback interface 'lo' and an ethernet interface 'eth0'. The 'arp -a' command is executed, displaying the ARP table with entries for the loopback interface and the ethernet interface.

```
(kali@kali)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:00:27:ba:84:f2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.12/24 brd 192.168.1.255 scope global dynamic noprefixroute eth0
        valid_lft 85102sec preferred_lft 85102sec
    inet6 2a02:a00:1409:d15:50fb:1a47:b002:7209/64 scope global dynamic noprefixroute
        valid_lft 86239sec preferred_lft 3429sec
    inet6 fe80::d816:9900:7a05:d30e/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(kali@kali)-[~]
$ arp -a
LAPTOP-TCDKMG.bbrouter (192.168.1.6) at 00:e2:50:93:18:e5 [ether] on eth0
HPK_QW.bbrouter (192.168.1.1) at 60:31:92:16:8c:a5 [ether] on eth0
COOLITA.bbrouter (192.168.1.7) at 88:28:7d:21:ab:9f [ether] on eth0
VX311.bbrouter (192.168.1.9) at aa:4b:20:01:3d:c1cf [ether] on eth0

(kali@kali)-[~]
$
```

3. Perform a port scan on your own computer using Nmap to find which common ports (like 22, 80, 443) are open. Summarize your findings in a table with port number, service, and status (open/closed).



The terminal shows the execution of an Nmap scan on the local host (192.168.1.12). The output indicates that the host is up and lists the open ports: 22/tcp (ssh), 80/tcp (http), and 443/tcp (https).

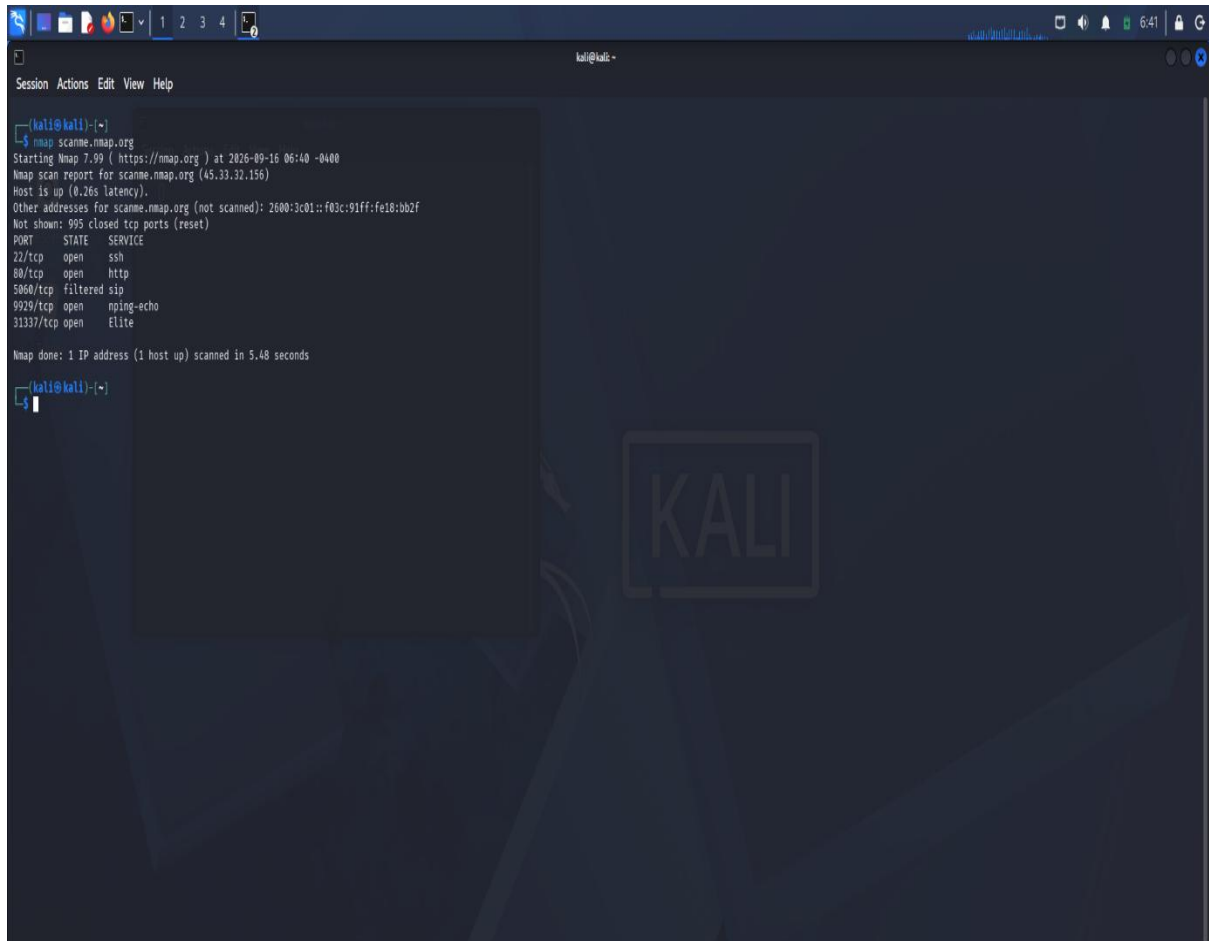
```
(kali@kali)-[~]
$ nmap -sV -p 22,80,443 192.168.1.12
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-16 06:35 -0400
Nmap scan report for kali.bbrouter (192.168.1.12)
Host is up.

PORT      STATE SERVICE VERSION
22/tcp    filtered ssh
80/tcp    filtered http
443/tcp   filtered https

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 3.23 seconds

(kali@kali)-[~]
$
```

4. **Simulate a Zomato-style scenario: Imagine you are a security tester for a food delivery app. Use Nmap to check if the test server at 'scanme.nmap.org' has any open ports that could be vulnerable. List any ports found and suggest what kind of services might be running on them.**



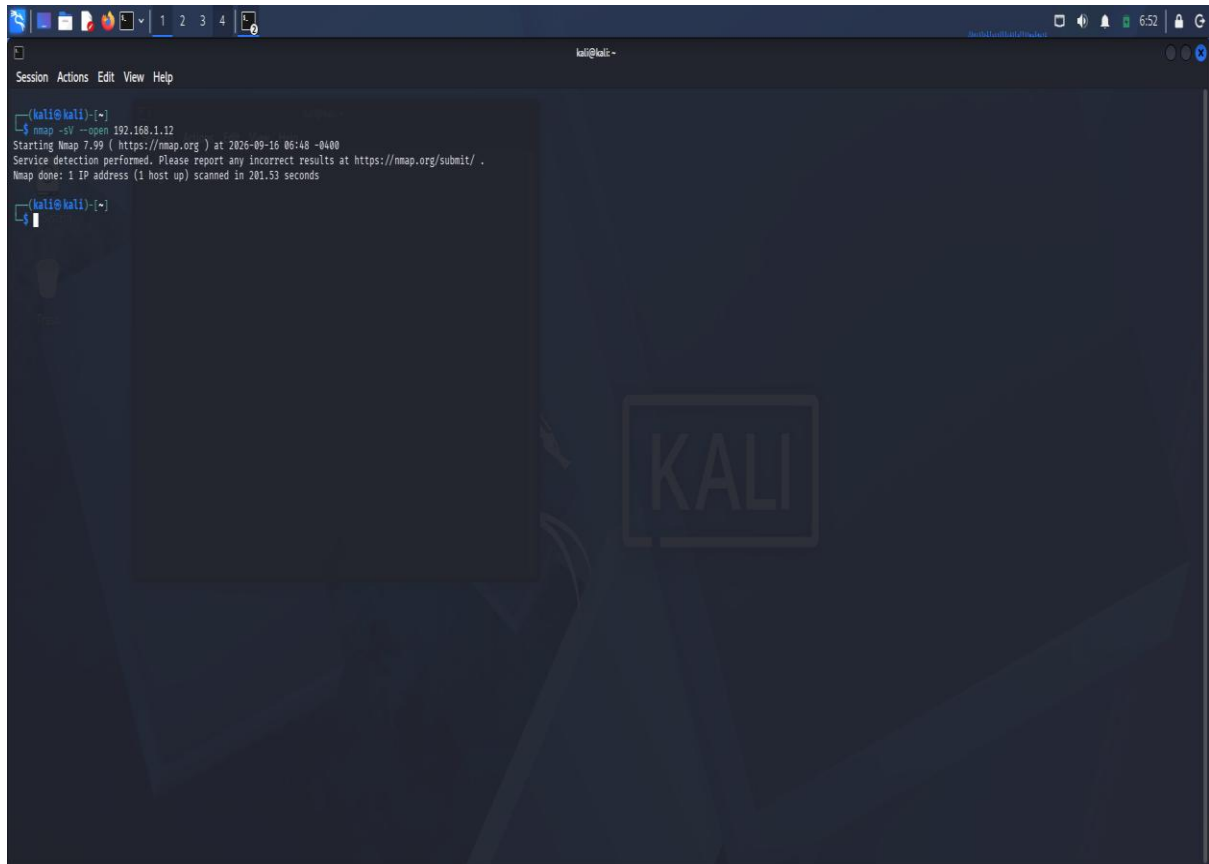
```
(kali@kali)~$ nmap scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-16 06:40 -0400
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.26s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 995 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
5060/tcp   filtered sip
9929/tcp   open  nping-echo
31337/tcp  open  Elite

Nmap done: 1 IP address (1 host up) scanned in 5.48 seconds

(kali@kali)~$
```

Nmap was used to scan the authorized test server scanme.nmap.org. This scan identified open ports 22, 80, 9929 and 31337 along with filtered port 5060. These ports indicate services such as SSH, HTTP and network diagnostic services may be running. Open ports alone do not confirm vulnerabilities. So, further authorized service-version analysis would be required.

5. Refactor your Nmap scan command to only show open ports and their services, hiding all closed ports from the output.



The screenshot shows a Kali Linux terminal window with a dark blue background and a faint 'KALI' logo. The terminal displays the following text:

```
kali@kali:~$ nmap -sV -oN 192.168.1.12
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-16 06:48 -0400
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 201.53 seconds

kali@kali:~$
```